KOPI

INTERNAL REPORT

USER MANUAL FOR THE
COMPUTER PROGRAM CARL

52406-45 10th June, 1985

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RUN DESCRIPTION

The CARL-program can be found on NGI's PRIME computer, using:

SEG PROGLIB>CARL

The program will ask for the names of the input data file and the output listing file.

The program will automatically open and close the necessary working files used.

(IMPORTANT: Print option NPR4 = 1 should always be used in the first runs to map possible minimum-areas for the safety factor, before using the simplified searching routine NPR4 = 0)

DESCRIPTION OF INPUT TO CARL

The standard input form is shown on drawing 52406-44-1.

1. LINE (PROJECT) FORMAT 14A4, A1

   01-57 - Heading of run, project, case, operators name etc.

2. LINE (UNITS USED) FORMAT 7A4, A2

   01-30 - Length and force units used. Any set of units may be chosen, but it is important that input values are given with consistent units. Output units will be equal to input units.

3. LINE (PLATFORM BASE GEOMETRY, ETC) B, AL, ZSS, ZSF, ZSSTEP, GAMS; GAMW, PS FORMAT 8F10.0

01-10 B	- Width of foundation, measured parallel to the direction of the horizontal force.
11-20 AL	- Length of foundation, measured normal to the direction of the horizontal force.
21-30 ZSS	- Depth of embedment, start value for ZS.
31-40 ZSF	- Depth of embedment, final value for ZS.
41-50 ZSSTEP	- Depth of embedment, step value for ZS. If calculations are performed for one value only, ZSSTEP = 0.0 and ZSS = ZSF. Maximum number of ZS-values is 100.
51-60 GAMS	- Total unit weight of soil (Force/length^3)
61-70 GAMW	- Unit weight of water (Force/length^3)
71-80 PS	- Selection of parameter study. $0 < PS < 1.0$: A selection of variables will be increased and decreased by the percentage indicated by PS (PS = 0.1 means a variation of ± 10%). These variables are: Area, (both B and AL are varied), depth of embedment (only one value should be given, see column 21-50 above), vertical load, horizontal load, moment, excess water pressure and load factor. Only one value of submerged weight is used, PVS. The only output available for parameter study is "summary of results". If a regular run is wanted, PS should be left blank.

4. LINE (FORCES ON PLATFORM) PVS, PVF, PVSTEP, PH, AM, DELPH, DELPT, FAC FORMAT 8F10.0

01-10 PVS	- Submerged weight of structure, start value. Includes weight of structure above mudline and possible environmental vertical forces

on structure.

11-20 PVF - Submerged weight of structure, final value.

21-30 PVSTEP - Submerged weight of structure, step value. If calculations are performed for one value only, PVSTEP = 0 and PVS = PVF. Maximum number of P_V-values is 20.

31-40 PH - Horizontal load at mudline. Value is disregarded if NPR5 = 1.

41-50 AM - Overturning moment at mudline

51-60 DELPH - Excess water pressure on sea floor adjacent to platform outside the heel, (loward side).

61-70 DELPT - Excess water pressure on sea floor adjacent to platform outside the toe, (leeward side).

71-80 FAC - Load factor(s) to be used for horizontal force (PH), moment (AM) and excess water pressure (DELPH, DELPT). Three options are available:

FAC = 1.0,	only loadfactor 1.0 will be used
FAC > + 1.0,	loadfactor 1.0 <u>and</u> the given value of FAC will be used
FAC < - 1.0,	only one loadfactor, $\lvert FAC \rvert$ will be used

5. LINE (SLIP SURFACE PARAMETERS) ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, ZSPEC1, ZSPEC2 FORMAT 8F10.0

01-10 ZMAX - The maximum depth from mudline to which the program should search for the minimum safety factor. The program will not accept any value of ZMAX greater than (ZS + BH). If Z exceeds this value the safety factor will be given the value of 99.9. The number of horizontal locations of the circle centre is automatically set to 11.

11-20 ZSTEP - The depth Z to the lowest point on the slip surface is varied in steps. The first depth is (ZS+ZSTEP) and the last ZMAX. Maximum 100 steps can be used.

21-30 REDBS - Reduction factor for side shear on embedded base area. The horizontal resistance of the side base area is calculated as the in-situ shear strength of the soil times the side area times REDBS. Due to soil disturbance during the penetration phase, REDBS is normally chosen as 0.5.

31-40 REDSS - Reduction factor for the resistance of the soil side areas, which is calculated as the in-situ shear strength of the soil times soil side area times REDSS. A factor of 0.4 has been proven to give results in good agreement with Brinch Hansen's formula for homogeneous soil conditions in the CAP program, and this value is therefore temporarily recommended.

41-50 CRACK - The active earth pressure is not allowed to be negative. In such case an open crack should be assumed between the skirt and the soil.
CRACK = 0.0 means open crack.
CRACK = 1.0 means closed crack.

51-60 PASTYP - The opportunity to choose between two types of passive failure zones is not present in this program, use 0.0, value is disregarded.

61-70 ZSPEC1 - Option for extra output concerning a specified failure surface. ZSPEC1 is the specified BH-value, i.e. the horizontal distance from circle center to right edge. If ZSPEC2 is specified greater than zero, then ZSPEC1 also has to be assigned a value.

71-80 ZSPEC2 - Specified UH-value, shear circle depth below right edge of base. If ZSPEC1 is specified greater than zero, then ZSPEC2 also has to be assigned a value.

6. LINE (PRINT OPTIONS, AREA VARIATIONS ETC.) NSU, NPR1, NPR2 NPR3, NPR4, NPR5, BMAX, DELB, ALMAX, DELAL, RUV, RUH, BHMIN. FORMAT F5.0, 5F3.0, 4F10.0,2F5.2,F10.0. (The first six variables may be given as integers, right justified).

01-05 NSU - total number of shear strength input data lines that follows, i.e. total number of line 7. NSU $\leq$ 50.

06-08 NPR1 - print option for output of table with calculated safety factor versus depth of circle (ZS+UH) and location of circle centre (BH)

NPR1 = 1; tables are given for each combination of ZS, PV and LF

NPR1 = 0; tables are not given.

Notice that complete table is only given if NPR4 = 1.

09-11 NPR2 - print option for output of table with all calculated moments for the critical slip surface

NPR2 = 1; tables are given for each combination of ZS, PV and LF.

NPR2 = 0; tables are not given.

12-14 NPR3 - print option which is dummy in CARL. The number will be read but not used. If ZSPEC1 and ZSPEC2 is greater than zero NPR3 will automatically be set to 1 in the program.

15-17 NPR4 - print option to choose between calculation of all safety factors (NZ • 11) or calculation according to a simplified searching routine.

NPR4 = 0; searching routine

NPR4 = 1; calculation of all the safety

factors.

18-20 NPR5 - option to calculate minimum horizontal force when SF = 1.0.
NPR5 = 0; searching for minimum safety factor (PH = fixed)
NPR5 = 1; searching for minimum horizontal force (SF = 1.0)
If NPR5 = 1 then NPR1 = NPR2 = NPR3 = NPR4 = 0 are fixed, and only loadfactor 1.0 will be considered.

21-30 BMAX - Area may be changed in steps from (B • AL)
31-40 DELB up to (BMAX • ALMAX). DELB and DELAL are
41-50 ALMAX step values. If BMAX is reached before
51-60 DELAL ALMAX, BMAX is kept unchanged until ALMAX is reached, and vice versa.
Each area (B(n) • AL(n)) is treated as a separate run with full output; that is "Data input reprint, Tables required and Summary of results".
If no area change is wanted, these four fields should be left blank.

61-65 RUV - roughness factor on left side (active earth pr.)

66-70 RUH - roughness factor on right side (passive earth pr.)

71-80 BHMIN - minimum horizontal distance from circle center to right edge. BHMIN has to be specified greater than zero, and preferably ⩾ B/2.

7. LINE (SHEAR STRENGTH INPUT DATA) ZZZ, SU, SUD, SUP, SU2
Maximum 50 lines, FORMAT 5F10.0

01-10 ZZZ - The depth from mudline at which the shear strength values are given. The first line should always be given for ZZZ = 0.0.
If ZMAX is greater than ZZZ (NSU) the number NSU will be increased by 1. If NSU has already been specified as 50, this will cause

error. The shear strength value at depth ZMAX will then be the same as ZZZ(NSU). The shear strength values between the specified depths will be found by linear interpolation.

11-20 SU - Soil shear strength determined by undrained triaxial compression tests, or just ordinary undrained shear strength in the case of a simple total stress analysis.

21-30 SUD - Soil shear strength determined by undrained simple shear tests. In the case of a simple total stress analysis, this field should be left blank. If values of SUD are to be given, the value at mudline must be > 0.0, and values have to be given in the whole table. If omitted, the values of SUD will be set equal to SU.

31-40 SUP - Soil shear strength determined by undrained triaxial extension tests. Further comments and limitations as for SUD.

41-50 SU2 - Minimum undrained shear strength is DUMMY in CARL. Should be left blank.

DIVIDING DATA SETS

01-04 **** Between different data sets in the same run, a line with **** in columns 1-4 has to be inserted. A "data set" consist of LINE 1 to 6 plus NSU LINE 7.

END OF LAST DATA SET

At the end of the last data set a blank line has to be inserted.

PROGRAM "FLOW CHART" FOR CARL

```
MAIN2
     INPUT2
     CALC1
            ARRAY
            RPRINT
            FACT
            SAF2
                    SHEAR2
                             SUBS
                    FORCE
            FORCC2
                    HHEAD
                    SHEAR2
                             SUBS
                    FORCE
                    HEAD
                    HANSEN
                    MEYER
            FORCE2
                    SHEAR2
                             SUBS
                    FORCE
                    HEAD
            OUTPUT
     CALC2
            as CALC1 plus:
            SAF1
                     SAF2
                             SHEAR2
                                     SUBS
                             FORCE
    OUTPUT
    CALC5   ARRAY
            RPRINT
            FACT
            HORF2
                     SHEARZ
                             SUBS
                     FORCE
            HORF1
                     HORF2
                             SHEAR2
                                      SUBS
                             FORCE
            OUTPUT5
    OUTPUT5
```

SUBROUTINES

The stability program CARL consists of the following subroutines:

ARRAY - Generates the necessary number of each variable for which calculations should be carried out.
INPUT: XF, XL, XS
OUTPUT: X, N

CALC2 - Organizes all the calculations to be carried out if NPR4 = 0 and NPR5 = 0.
INPUT: IPAGE, ITEXT(30), BB, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS, PVF, PVSTEP, NLOAD, ALF(2), ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, NSU, NPR1, NPR2 NPR3, ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), PHO, AMO, DELPHO, DELPTO, NSPEC, ZSPEC(3), NUM, ALL(50), BBB(50), PS, NPS, IPROG, IUNIT(16), RUV, RUH, BHMIN
OUTPUT: NPV, NZS
Both output parameters are generated in subroutine ARRAY
INPUT/OUTPUT: IPAGE
This is changed in subroutine FORCC2 and FORCE2
EXTERNAL CALLS: ARRAY
RPRINT
FACT
SAF2
FORCC2
FORCE2
OUTPUT

FACT - Generates several factors to be used in the calculation of forces.

INPUT: REDBS, REDSS, CRACK.
OUTPUT: F(16)

FORCE - Calculates all the A- and B-values in the general equations for the forces.
INPUT: PV, ZS, BB, AL, UH, BH, PH, AM, DELPH, DELPT, GAMS, GAMW, RUV, RUH, F, SA
OUTPUT: A(20), B(20)

FORCC2 - Calculates safety factor, forces and mobilized shear stresses for critical slip surface
INPUT: PV, ZS, ALF2, NZ, Z(100), SFMIN(11,100) SUMB, IPAGE, ITEXT(30), BB, BHMIN, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS, PVF, PVSTEP, PH, AM, DELPH, DELPT, NLOAD, ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), F, RUN, RUH, NSU, NPR1, NPR2, NUM
INPUT/OUTPUT: IPAGE
being updated during this routinge
OUTPUT ON TAPE: ALF, PV, ZS, SF, Z(100), BHC, CSS, CB, CM, I22, I33
EXTERNAL CALLS: HHEAD
SHEAR2
FORCE
HEAD
HANSEN
MEYER

FORCE2 - Calculates safety factor, forces and mobilized shear stresses for specified slip surfaces.
INPUT: PV, ZS, ALF2, IPAGE, ITEXT(30), BB, AM, AL, ZSSTEP, PVSTEP, PH, DELPH, DELPT, NPR1, NPR2, PVS, PVF, ZSF, ZSS, NLOAD, SU(50), SUD(50), SUP(50), SU2(50), GAMS, GAMW, F, RUV, RUH, NSPEC, ZZZ(50), NSU, NUM, ZSPEC(3)
INPUT/OUTPUT: IPAGE, NPAGE
EXTERNAL CALLS: HHEAD, SHEAR2, FORCE, HEAD

HANSEN - Calculates mobilized shear stress according to Brinch Hansens bearing capacity formula modified for offshore structures.

INPUT: BB, AL, D, GAM, GAMW, PH, AMM, DPH, DPT, PV

OUTPUT: C3, I2

HEAD - Provides the writing of a general heading

INPUT: NPAGE, IPAGE, ITEXT(30), ALF2, PV, ZS

HHEAD - Provides the writing of a general heading of the critical slip surface.

INPUT: NPAGE, IPAGE, ITEXT(30), ALF2, PV, ZS.

INPUT2 - Reads all necessary data from input file Provides the writing of "Input data reprint" and makes all data ready for processing in the different subroutines.

OUTPUT: IPAGE, ITEXT(30), BB, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS, PVF, PVSTEP, ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, NSU, NPR1, NPR2, NPR3, NPR4, NPR5, SU(50), SUD(50), SUP(50), SU2(50) NSPEC, ALL, BBB, PS, IPROG, IUNIT, RUV, RUH, BHMIN, ZSPEC, ALF, NLOAD, PHO, AMO, DELPHO, DELPTO, ZZZ(50), NUM.

MEYER - Calculates mobilized shear stress according to Meyerhof's bearing capacity formula modified for offshore structures.

INPUT: BB, AL, D, GAM, GAMW, PH, AMM, DPH, DPT, PV

OUTPUT: C2, I2

OUTPUT - Provides the writing of "Summary of results" - table.

INPUT: IPAGE, ITEXT(30), ZSS, ZSF, ZSSTEP, PVS, PVSTEP, NLOAD, NPR1, NPR2, NPV,

NZS, NUM, PS, NPS

INPUT/OUTPUT: IPAGE

being updated during this routine

RPRINT - Provides the writing of all the variables in use at any time.

INPUT: BB, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS, PVSTEP, PH, AM, DELPH, DELPT, FAC, ZMAX, ZSTEP, REDBS, REDSS, IUNIT(16), ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), NSU, NLOAD, NPR1, NPR2, NUM, IPROG, IPAGE, ITEXT(30), RUV, RUH, BHMIN, CRACK

OUTPUT: IPAGE

being updated during this routine.

SAF2 - Calculates the safety factor for a given data set.

INPUT: PV, ZS, BB, AL, UH, BH, PH, AM, DELPH, DELPT, NSU, ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), GAMS, GAMW, RUV, RUH, F.

OUTPUT: SF, CSS, SUMB

EXTERNAL CALLS: SHEAR2, FORCE

SHEAR2 - Calculates all the average shear strength values to be used in the general equations for the forces.

INPUT: BB, BH, ZS, UH, NSU, ZZZ(50), SU(50),SUD(50), SUP(50), SU2(50)

OUTPUT: SA(20)

INPUT/OUTPUT: F(16) is changed during this routine

EXTERNAL CALL: SUBS

SUBS - Calculates actual shear strength value for given depth by interpolation.

INPUT: ZZZ(50), SU(50), NSU, Z

OUTPUT: SS

CIRCLE - Calculates the geometry of the CARL-circle
INPUT: BH, UH, BB
OUTPUT: BEV, BEH, BV, UV, R.

CALC1 - Organizes all the calculations to be carried out if NPR4 = 1 and NPR5 = 0.
INPUT: IPAGE, ITEXT(30), BB, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS, PVF, PVSTEP, NLOAD, ALF(2), ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, NSU, NPR1, NPR2, NPR3, ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), PHO, AMO, DELPHO, DELPTHO, NSPEC, ZSPEC(3), NUM, ALL(50), BBB(50), PS, NPS, IPROG, IUNIT(16), RUV, RUH, BHMIN.
Output: NPV, NZS
Both output parameters are generated in subroutine ARRAY
INPUT/OUTPUT: IPAGE
This is changed in subroutine FORCCE2 and FORCE2
EXTERNAL CALLS: ARRAY
RPRING
FACT
SAF2
FORCC2
FORCE2
OUTPUT

SAF1 - checkes if BH and UH are within allowable range. If so it calls SAF2 and finds the safety factor.
INPUT: K2, N2, PVJ, SK, BB, AL, Z(100), BHMIN, PH, AM, DELPH, DELPT, NSU, ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), GAMS, GAMW, RUV, RUA, F(16), NZ
OUTPUT: SF MIN(11, 200) CSS, SUMB(11, 200)

EXTERNAL CALL: SAF2

CALC5 - Organizes all the calculations to be carried out, if NPR4 = 0 and NPR5 = 1

INPUT: IPAGE, ITEXT(30), BB, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS, PVF, PVSTEP, NLOAD, ALF(2), ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, NSU, NPR1, NPR2, NPR3, ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), PHO, AMO, DELPHO, DELPTO, NSPEC, ZSPEC(3), NUM, ALL(50), BBB(50), PS, NPS, IPROG, IUNIT(16, RUV, RUH, BHMIN

OUTPUT: NPV, NZS

Both output parameters are generated in subroutine ARRAY

INPUT/OUTPUT: IPAGE

EXTERNAL CALLS: ARRAY
RPRING
FACT
OUTPUT5
HORF1
HORF2

HORF1 - checks if BH and UH are within its allowable range. If so it calls HORF2 and finds the horizontal force.

INPUT: K2, N2, PVJ, ZK, BB, AL, Z(100), BHMIN, AM, DELPH, DELPT, NZ, NSU, ZZZ(50), SU(50), SUD(50), SUP(50), SU2(50), GAMS, GAMW, RUV, RUH, F(16)

OUTPUT: PHMIN(11, 200), CSS, SUMB(11, 200)

EXTERNAL CALL: HORF2

HORF2 - calculates the horizontal force for a given dataset included safety factor equal to 10.

INPUT: PV, ZS, BB, AL, UH, BH, AM, DELPH, DELPT, NSU, ZZZ(50), SU(50), SUP(50), SUP(50), SU2(50), GAMS,

GAMW, RUV, RUH, F(16)

OUTPUT: PH, CSS, SUMB

EXTERNAL CALLS: SHEAR2, FORCE

OUTPUT5 - Provides the writing of "summary of results" - table if NPR5 = 1

INPUT: IPAGE, ITEXT, ZSS, ZSF, ZSSTEP, PVS, PVF, PVSTEP, NLOAD, NPR1, NPR2, NPV, NZS, NUM, PS, NPS)

NGI

ALPHABETIC LIST OF SYMBOLS USED IN CARL

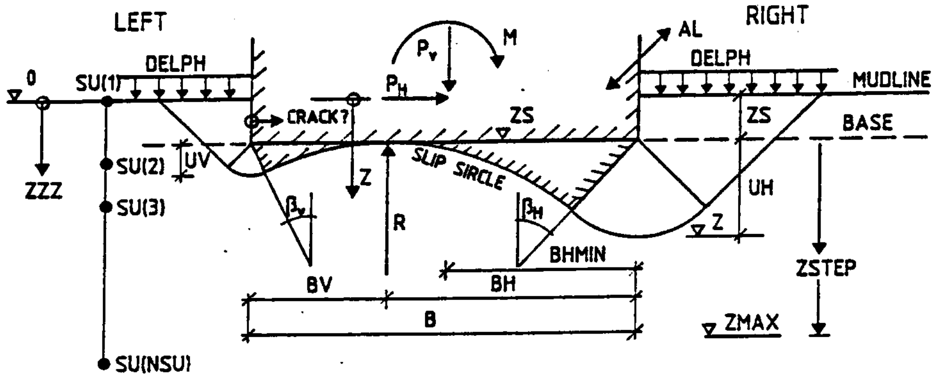
A(20)	=	factor in general force expression
AK(20)	=	moment around circle centre
AL	=	base length normal to wave direction
ALF(2)	=	load factor
ALF2	=	load factor
ALL(50)	=	AL
AM	=	moment at sea floor
AMM	=	AM
AMO	=	start value of AM
B(20)	=	factor in general force expression
B	=	base width in wave direction
BBB(50)	=	BB, array of B-values
BO	=	effective base width
CB	=	mobilized shear stress according to Brinch Hansen's formula
CM	=	mobilized shear stress according to Meyerhof's formula
CRACK	=	option for crack on active side
CSS	=	mobilized shear stress according to CARL
C2	=	CM
C3	=	CB
D	=	depth of embedment
DELPH	=	excess wave pressure at sea floor outside heel
DELPT	=	excess wave pressure at sea floor outside toe
DELPHO	=	start value for DELPH
DELPTO	=	start value for DELPT
DPH	=	DELPH
DPT	=	DELPT
E	=	eccentricity

F(16)	=	factors used for calculating forces
FAC(2)	=	load factor
GAM	=	total unit weight of soil
GAMS	=	total unit weight of soil
GAMW	=	unit weight of water
IPAGE	=	page number
ITEXT(30)	=	array for storage of text from first line of input
IUNIT(16)	=	array for storage of text from second line of input
N	=	number of values
NLOAD	=	number of load factors
NPAGE	=	number of pages in output
NPR1	=	print option 1
NPR2	=	print option 2
NPR3	=	print option 3 (dummy)
NPR4	=	print option 4
NPR5	=	option 5
NPS	=	15 = number of cases in parameter study
NPV	=	number of vertical loads
NSPEC	=	number of specified depths
NSU	=	number of shear strength values
NUM	=	number of loops for each variable
NZ	=	number of depths
NZS	=	number of depths of embedment
PASTYP	=	print option for passive failure zone (dummy)
PH	=	horizontal load at sea floor
PHMIN(11,200)	=	array containing all PH values if NPR5 = 1
PHO	=	start value of PH
PS	=	selection of parameter study
PV(20)	=	vertical load at sea floor
PVF	=	vertical load at sea floor, final value
PVS	=	vertical loadd at sea floor, start value
PVSTEP	=	vertical load at sea floor, step value

REDBS	=	reduction factor base/soil
REDSS	=	reduction factor soil/soil
RUH	=	roughness right side
RUN	=	roughness left side
SA(20)	=	average shear strength
SF	=	safety factor
SFMIN(11,100)	=	all the safety factors
SS	=	value of shear strength at specified depth
SU(50)	=	undrained shear strength, by compression test
SUD(50)	=	undrained shear strength, by simple shear test
SUP(50)	=	undrained shear strength, by extension test
SU2(50)	=	undrained shear strength, minimum
SUMB(11,100)	=	all the sums of the B-values
Z	=	depth below sea floor
ZMAX	=	maximum depth for the program to search
ZS(100)	=	depth of embedment
ZSF	=	depth of embedment, final value
ZSS	=	depth of embedment, start value
ZSSTEP	=	depth of embedment, step value
ZSPEC(2)	=	specified BH value
ZSPEC(3)	=	specified UH value
ZSTEP	=	step value during the search of SFMIN
ZZZ(50)	=	Z-value corresponding to the shear strength values
X	=	variable
XF	=	variable, first value
XL	=	variable, last value
XS	=	variable, step value

REVISED INPUT
DATA FORM
FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



1 – 57
PROJECT

1 – 30
UNITS USED

1–10	11–20	21–30	31–40	41–50	51–60	61–70	71–80
B	AL	ZSS	ZSF	ZSSTEP	GAMS 1)	GAMW 1)	PS *)

1–10	11–20	21–30	31–40	41–50	51–60	61–70	71–80
PVS	PVF	PVSTEP	PH	AM	DELPH	DELPT	FAC 2)

1–10	11–20	21–30	31–40	41–50	51–60	61–70	71–80
ZMAX	ZSTEP	REDBS 3)	REDSS 4)	CRACK 9)	PASTYP	ZSPEC 1 *)	ZSPEC 2 *)
					////		

1–4	5–7	8–10	11–13	14–16	17–20	21–30	31–40	41–50	51–60	61–65	66–70	71–80
NSU	NPR1 8)	NPR2 8)	NPR3 8)	NPR4 8)	NPR5 8)	BMAX *)	DELB *)	ALMAX *)	DELAL *)	RUV 10)	RUH 10)	BHMIN
			////									

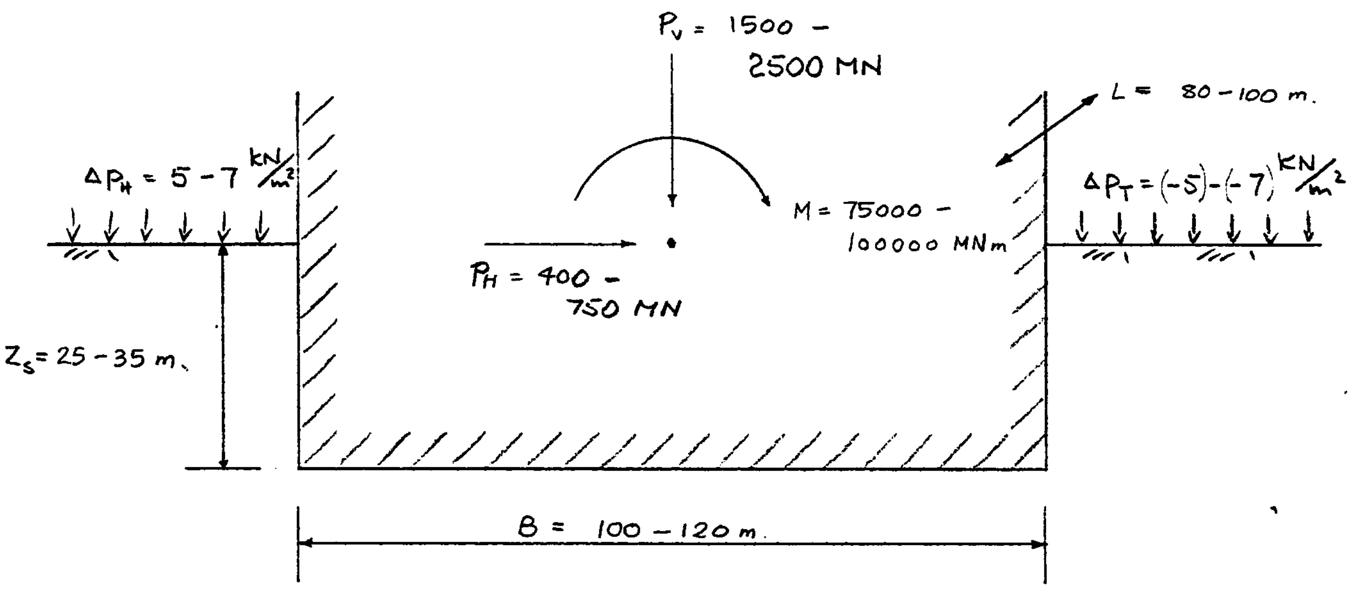
	1–10	11–20	21–30	31–40	41–50
	ZZZ	SU 5)	SUD 6)	SUP 7)	SU2
1					////
2					////
3					////
4					////
5					////
6					////
7					////
8					////
9					////
10					////
11					////
12					////
13					////
14					////
15					////

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
= 1.2, LOAD FACTORS 1.0 AND 1.2
= -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 ≠ 0 GIVES TABLE OF SF-VALUES
NPR2 ≠ 0 GIVES ALL K (N) FOR SFMIN
NPR3 = 0 OMITTED IN THIS MODE
NPR4 = 0 SIMPLIFIED SEARCHING ROUTINE
" ≠ 0 COMPLETE TABLE OF SF-VALUES
NPR5 ≠ 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE. = 0. OPEN CRACK.: = 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
*) ZSPEC1 = BH, ZSPEC2 = UH.

*) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
BLANK LINE ENDING ALL DATA SETS.

APPENDIX II : EXAMPLES

Summary of data variations :



Skj 013E 100 000. 5-84. BRA-TRYKK A/S

Norges Geotekniske Institutt

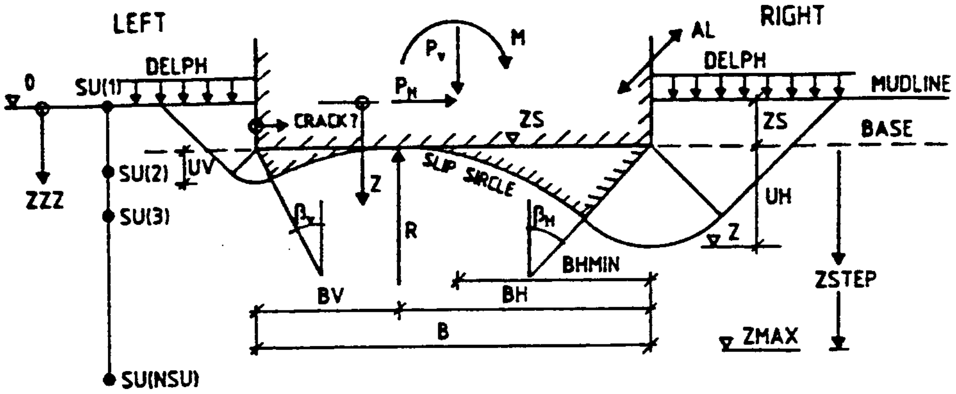
OPPDRAG: CAP-FAMILY EXAMPLES	OPPDRAGSNR. 52406-44	SIDE II - 2
FORSØK / BEREGNING: FUNCTION CONTROL	SIGN. RAL	DATO 24/5-85
	KONTR.	DATO

Function to be checked in		RUN NO: 1	2	3	4	5	6	7
1. ZSTEP,	calculation for z_s-values		x	x				
2. GAMS,	calculation independant of γ-value or not ?			x				
3. PS,	parameter study	x						
4. PVSTEP,	calculation for P_v-values			x				
5. FAC,	1.0 only $\gamma_f = 1.0$	x			x	x	x	x
	1.3 both $\gamma_f = 1.3$ and 1.0		x					
	-1.3 only $\gamma_f = 1.3$			x				
6. REDBS,	reduction base/soil	.5	.5	.5	0	.5	1.	.5
7. REDSS,	" soil/soil	.4	.4	.4	0	.4	1.	.4
8. CRACK,	0.0 open crack				x			
	1.0 closed crack	x	x	x		x	x	x
9. PASTYP,	0.0 or 1.0, difference ?	1.	1.	0.	0.	1.	1.	1.
10. ZSPEC1 & 2,	Specified failure surface, K(N) table					x		
11. NPR 1,	0, 1 Table SF/z	0	1	0	1	1	0	1
12. NPR 2,	0, 1 Table K(N) for SF_{min}	0	0	0	1	1	1	1
13. NPR 4,	0, 1 Full table SF/z	0	0	0	0	1	0	1
14. NPR 5,	0, 1 P_H for SF = 1.0	0	0	0	0	0	0	1
15. BMAX/DELB,	more areas in one run						x	
16. RUV/RUH,	earth pressure coefficients	.5	.5	.5	0	.5	1.0	.5
17. SU,	shear strength options		x					
18. SU/SU2,	- " -			x				
19. SU, SUD, SUP,	- " -	x						
20. SU, SUD, SUP, SU2,	- " -							x
21. More datasets,	in one run.				x			

SKJ 013N. 100 000 8-84 BRA-TRYKK A/S

STANDARD INPUT
DATA FORM
FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



PROJECT (1–57)
EXAMPLE 1 – CARL

UNITS USED (1–30)
MN/M

B (1–10)	AL (11–20)	ZSS (21–30)	ZSF (31–40)	ZSSTEP (41–50)	GAMS 1) (51–60)	GAMW 1) (61–70)	PS *) (71–80)
100.0	100.0	30.0	30.0	0.0	0.02	0.01	0.1

PVS (1–10)	PVF (11–20)	PVSTEP (21–30)	PH (31–40)	AM (41–50)	DELPH (51–60)	DELPT (61–70)	FAC 2) (71–80)
1500.0	1500.0	0.0	400.0	75000.0	0.005	-0.005	1.0

ZMAX (1–10)	ZSTEP (11–20)	REDBS 3) (21–30)	REDSS 4) (31–40)	CRACK 9) (41–50)	PASTYP (51–60)	ZSPEC 1 *) (61–70)	ZSPEC 2 *) (71–80)
70.0	2.0	0.5	0.4	1.0	1.0		

NSU (1–5)	NPR1 8) (8)	NPR2 8) (11)	NPR3 8) (14)	NPR4 8) (17)	NPR5 8) (20)	BMAX *) (21–30)	DELB *) (31–40)	ALMAX *) (41–50)	DELAL *) (51–60)	RUV 10) (61–65)	RUH 10) (66–70)	BHMIN (71–80)
2	0	0	0	0	0					0.5	0.5	50.0

	ZZZ (1–10)	SU 5) (11–20)	SUD 6) (21–30)	SUP 7) (31–40)	SU2 (41–50)
1	0.0	0.001	0.001	0.001	
2	100.0	0.225	0.200	0.180	
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
= 1.2, LOAD FACTORS 1.0 AND 1.2
= -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 = 0 GIVES TABLE OF SF-VALUES
NPR2 = 0 GIVES ALL K (N) FOR SFMIN
NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
NPR4 = 0 OMITTED IN THIS MODE
NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE. = 0. OPEN CRACK.:
= 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
*) ZSPEC1 = BH, ZSPEC2 = UH.

*) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
BLANK LINE ENDING ALL DATA SETS.

C A R L

------------------------------------ NGI 1984 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 01 12 1984

PROJECT : EKSEMPEL 1 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M		
WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
PARAMETERSTUDY	PS =	0.100
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.500
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK
PASSIVE SIDE AREA ASSUMPTION : BIG AREA

LEFT SIDE ROUGHNESS ASSUMTION : 0.50
RIGHT SIDE ROUGHNESS ASSUMTION : 0.50

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.200

PROJECT : EKSEMPEL 1 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

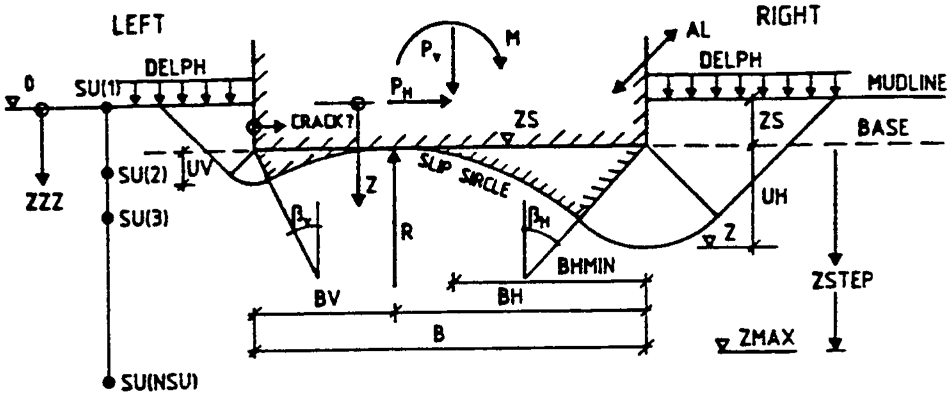
LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF		
LF	PV	ZS	SF	Z	BH	CSS	CB	CM		
1.00	1500.0	30.0	1.42	52.0	60.0	0.0389	0.0556	0.0573		BASIS
1.00	1500.0	30.0	1.09	52.0	54.0	0.0501	0.0747	0.0779	-0.10	B & L
1.00	1500.0	30.0	1.78	52.0	68.0	0.0310	0.0428	0.0439	0.10	B & L
1.00	1500.0	27.0	1.27	47.0	60.0	0.0395	0.0582	0.0601	-0.10	ZS
1.00	1500.0	33.0	1.56	55.0	60.0	0.0372	0.0531	0.0548	0.10	ZS
1.00	1350.0	30.0	1.43	52.0	60.0	0.0385	0.0540	0.0557	-0.10	PV
1.00	1650.0	30.0	1.40	54.0	65.0	0.0400	0.0571	0.0590	0.10	PV
1.00	1500.0	30.0	1.46	52.0	60.0	0.0378	0.0547	0.0566	-0.10	PH
1.00	1500.0	30.0	1.37	52.0	60.0	0.0401	0.0565	0.0582	0.10	PH
1.00	1500.0	30.0	1.50	52.0	65.0	0.0362	0.0516	0.0532	-0.10	M
1.00	1500.0	30.0	1.34	52.0	60.0	0.0411	0.0596	0.0618	0.10	M
1.00	1500.0	30.0	1.42	52.0	60.0	0.0388	0.0555	0.0572	-0.10	DELPH & DELPT
1.00	1500.0	30.0	1.41	52.0	60.0	0.0390	0.0557	0.0574	0.10	DELPH & DELPT
0.90	1500.0	30.0	1.55	54.0	65.0	0.0361	0.0507	0.0523	-0.10	LF
1.10	1500.0	30.0	1.30	52.0	60.0	0.0424	0.0607	0.0628	0.10	LF

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

NORWEGIAN GEOTECHNICAL INSTITUTE 52406-44-1

STANDARD INPUT
DATA FORM
FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



PROJECT (1–57)
EXAMPLE 2 – CARL

UNITS USED (1–30)
MN – M

B (1–10)	AL (11–20)	ZSS (21–30)	ZSF (31–40)	ZSSTEP (41–50)	GAMS 1) (51–60)	GAMW 1) (61–70)	PS *) (71–80)
100.0	100.0	25.0	35.0	5.0	0.02	0.01	

PVS (1–10)	PVF (11–20)	PVSTEP (21–30)	PH (31–40)	AM (41–50)	DELPH (51–60)	DELPT (61–70)	FAC 2) (71–80)
1500.0	1500.0	0.0	400.0	75000.0	0.005	-0.005	1.3

ZMAX (1–10)	ZSTEP (11–20)	REDBS 3) (21–30)	REDSS 4) (31–40)	CRACK 9) (41–50)	PASTYP (51–60)	ZSPEC 1 *) (61–70)	ZSPEC 2 *) (71–80)
70.0	2.0	0.5	0.4	1.0	1.0		

NSU (1–5)	NPR1 8) (8)	NPR2 8) (11)	NPR3 8) (14)	NPR4 8) (17)	NPR5 8) (20)	BMAX *) (21–30)	DELB *) (31–40)	ALMAX *) (41–50)	DELAL *) (51–60)	RUV 10) (61–65)	RUH 10) (66–70)	BHMIN (71–80)
2	1	0	0	0	0					0.5	0.5	50.0

	ZZZ (1–10)	SU 5) (11–20)	SUD 6) (21–30)	SUP 7) (31–40)	SU2 (41–50)
1	0.0	0.001			
2	100.0	0.225			
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
   = 1.2, LOAD FACTORS 1.0 AND 1.2
   = -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
   NPR1 = 0 GIVES TABLE OF SF-VALUES
   NPR2 = 0 GIVES ALL K (N) FOR SFMIN
   NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
   NPR4 = 0 OMITTED IN THIS MODE
   NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE = 0. OPEN CRACK.:
   = 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
*) ZSPEC1 = BH, ZSPEC2 = UH.

*) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
BLANK LINE ENDING ALL DATA SETS.

C A R L

------------------------------------ NGI 1985 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 10 6 1985

PROJECT : EKSEMPEL 2 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	25.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	35.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	5.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.3
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.500
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK
PASSIVE SIDE AREA ASSUMPTION : BIG AREA

LEFT SIDE ROUGHNESS ASSUMTION : 0.50
RIGHT SIDE ROUGHNESS ASSUMTION : 0.50

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 2 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 25.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH — HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE

DEPTH	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
27.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
29.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
31.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
33.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
35.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
37.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
39.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
41.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
43.0	0.00	1.34	1.32	1.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00
45.0	0.00	1.35	1.32	1.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00
47.0	0.00	1.36	1.32	1.32	1.35	1.39	1.44	0.00	0.00	0.00	0.00
49.0	0.00	0.00	1.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
51.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
53.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
55.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
57.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
59.0	0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.00	0.00	0.00	0.00
61.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
63.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
65.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
67.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

```
MINIMUM SAFETY FACTOR              SF =     1.316
FOUND FOR DEPTH                     Z =    45.000
AND DISTANCE TO CIRCLE CENTER      BH =    60.000
AND CIRCLE RADIUS                     =    80.000

S(A*SA)                               =-0.1784E+06
S(B)                                  = 0.1356E+06
```

PROJECT : EKSEMPEL 2 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
32.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
34.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
36.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
38.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
40.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
42.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
44.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
46.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
48.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
50.0	0.00	1.61	1.58	1.60	1.63	1.68	1.74	0.00	0.00	0.00	0.00
52.0	0.00	1.62	1.58	1.59	0.00	0.00	0.00	0.00	0.00	0.00	0.00
54.0	0.00	1.64	1.59	1.59	0.00	0.00	0.00	0.00	0.00	0.00	0.00
56.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
58.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
60.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
62.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
64.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
66.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
68.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MINIMUM SAFETY FACTOR SF = 1.582
FOUND FOR DEPTH Z = 52.000
AND DISTANCE TO CIRCLE CENTER BH = 60.000
AND CIRCLE RADIUS = 70.818

S(A*SA) =-0.2123E+06
S(B) = 0.1342E+06

PROJECT : EKSEMPEL 2 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 35.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
37.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
39.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
41.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
43.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
45.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
47.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
49.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
51.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
53.0	0.00	1.91	1.90	1.93	1.98	2.04	2.11	0.00	0.00	0.00	0.00
55.0	0.00	0.00	1.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
57.0	0.00	1.90	1.86	1.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00
59.0	0.00	1.91	1.86	1.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00
61.0	0.00	1.93	1.87	1.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00
63.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
65.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
67.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MINIMUM SAFETY FACTOR SF = 1.860
FOUND FOR DEPTH Z = 59.000
AND DISTANCE TO CIRCLE CENTER BH = 60.000
AND CIRCLE RADIUS = 63.000

S(A*SA) =-0.2480E+06
S(B) = 0.1333E+06

PROJECT : EKSEMPEL 2 ***CARL***
LOAD FACTOR : 1.30 SUBMERGED WEIGHT : 1500.0 BASE EMB : 25.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
27.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
29.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
31.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
33.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
35.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
37.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
39.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
41.0	0.00	1.04	1.04	1.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00
43.0	0.00	1.04	1.04	1.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00
45.0	0.00	1.05	1.04	1.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00
47.0	0.00	1.07	1.04	1.06	1.09	1.13	1.17	0.00	0.00	0.00	0.00
49.0	0.00	0.00	1.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
51.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
53.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
55.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
57.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
59.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
61.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
63.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
65.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
67.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MINIMUM SAFETY FACTOR SF = 1.038
FOUND FOR DEPTH Z = 43.000
AND DISTANCE TO CIRCLE CENTER BH = 60.000
AND CIRCLE RADIUS = 91.000

S(A*SA) =-0.1847E+06
S(B) = 0.1779E+06

PROJECT : EKSEMPEL 2 ***CARL***
LOAD FACTOR : 1.30 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
32.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
34.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
36.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
38.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
40.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
42.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
44.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
46.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
48.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
50.0	0.00	1.26	1.25	1.27	1.31	1.36	1.41	0.00	0.00	0.00	0.00
52.0	0.00	1.27	1.25	1.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00
54.0	0.00	1.28	1.25	1.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00
56.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
58.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
60.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
62.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
64.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
66.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
68.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MINIMUM SAFETY FACTOR SF = 1.249
FOUND FOR DEPTH Z = 52.000
AND DISTANCE TO CIRCLE CENTER BH = 60.000
AND CIRCLE RADIUS = 70.818

S(A*SA) =-0.2123E+06
S(B) = 0.1700E+06

PROJECT : EKSEMPEL 2 ***CARL***
LOAD FACTOR : 1.30 SUBMERGED WEIGHT : 1500.0 BASE EMB : 35.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
37.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
39.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
41.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
43.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
45.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
47.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
49.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
51.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
53.0	1.53	1.49	1.50	1.53	1.58	1.64	1.70	0.00	0.00	0.00	0.00
55.0	0.00	1.48	1.48	1.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00
57.0	0.00	1.48	1.47	1.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00
59.0	0.00	1.49	1.47	1.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00
61.0	0.00	1.50	1.47	1.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00
63.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
65.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
67.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MINIMUM SAFETY FACTOR	SF =	1.469
FOUND FOR DEPTH	Z =	59.000
AND DISTANCE TO CIRCLE CENTER	BH =	60.000
AND CIRCLE RADIUS	=	63.000

S(A*SA)	=-0.2480E+06
S(B)	= 0.1688E+06

PROJECT : EKSEMPEL 2 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

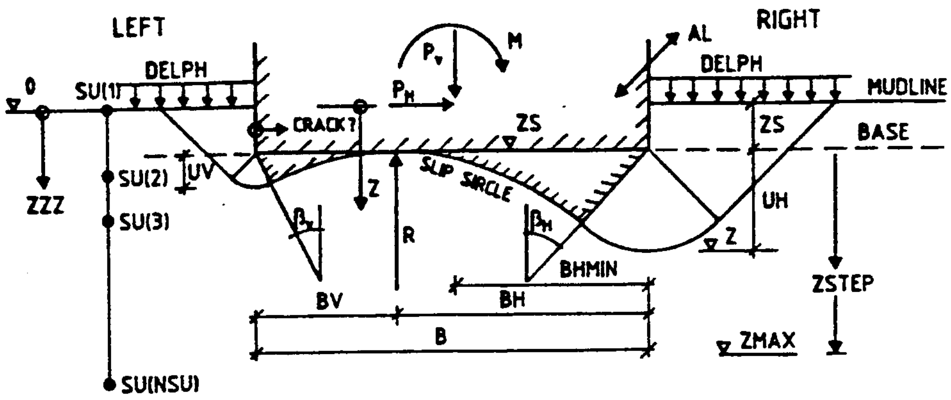
AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.00	1500.0	25.0	1.32	45.00	60.0	0.0409	0.0601	0.0621
1.00	1500.0	30.0	1.58	52.00	60.0	0.0389	0.0556	0.0573
1.00	1500.0	35.0	1.86	59.00	60.0	0.0372	0.0515	0.0533
1.30	1500.0	25.0	1.04	43.00	60.0	0.0500	0.0799	0.0832
1.30	1500.0	30.0	1.25	52.00	60.0	0.0493	0.0723	0.0751
1.30	1500.0	35.0	1.47	59.00	60.0	0.0471	0.0661	0.0687

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

STANDARD INPUT
DATA FORM
FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



PROJECT (1–57)
EXAMPLE 3 – CARL

UNITS USED (1–30)
MN/M

B (1–10)	AL (11–20)	ZSS (21–30)	ZSF (31–40)	ZSSTEP (41–50)	GAMS 1) (51–60)	GAMW 1) (61–70)	PS *) (71–80)
100.0	100.0	25.0	30.0	5.0	0.015	0.010	

PVS (1–10)	PVF (11–20)	PVSTEP (21–30)	PH (31–40)	AM (41–50)	DELPH (51–60)	DELPT (61–70)	FAC 2) (71–80)
1500.0	2500.0	500.0	400.0	75000.0	0.005	-0.005	-1.3

ZMAX (1–10)	ZSTEP (11–20)	REDBS 3) (21–30)	REDSS 4) (31–40)	CRACK 9) (41–50)	PASTYP (51–60)	ZSPEC 1 *) (61–70)	ZSPEC 2 *) (71–80)
70.0	2.0	0.5	0.4	1.0	0.0		

NSU (1–5)	NPR1 8)	NPR2 8)	NPR3 8)	NPR4 8)	NPR5 8) (–20)	BMAX *) (21–30)	DELB *) (31–40)	ALMAX *) (41–50)	DELAL *) (51–60)	RUV 10)	RUH 10) (–70)	BHMIN (71–80)
2	0	0	0	0	0					0.5	0.5	50.0

	ZZZ (1–10)	SU 5) (11–20)	SUD 6) (21–30)	SUP 7) (31–40)	SU2 (41–50)
1	0.0	0.001			0.001
2	100.0	0.225			0.050
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
 = 1.2, LOAD FACTORS 1.0 AND 1.2
 = -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
 NPR1 = 0 GIVES TABLE OF SF-VALUES
 NPR2 = 0 GIVES ALL K (N) FOR SFMIN
 NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
 NPR4 = 0 OMITTED IN THIS MODE
 NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE = 0. OPEN CRACK.
 = 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
*) ZSPEC1 = BH, ZSPEC2 = UH.

*) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.

C A R L

----------------------------------- NGI 1985 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES PAGE 1

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 10 6 1985

PROJECT : EKSEMPEL 3 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	25.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	5.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.015
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	2500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	500.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	-1.3
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.500
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK
PASSIVE SIDE AREA ASSUMPTION : SMALL AREA

LEFT SIDE ROUGHNESS ASSUMTION : 0.50
RIGHT SIDE ROUGHNESS ASSUMTION : 0.50

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.050

PROJECT : EKSEMPEL 3 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.30	1500.0	25.0	1.04	43.00	60.0	0.0500	0.1017	0.1072
1.30	1500.0	30.0	1.25	52.00	60.0	0.0493	0.0888	0.0934
1.30	2000.0	25.0	1.01	45.00	60.0	0.0534	0.0957	0.1008
1.30	2000.0	30.0	1.21	52.00	60.0	0.0508	0.0871	0.0915
1.30	2500.0	25.0	0.98	45.00	65.0	0.0544	0.0954	0.0999
1.30	2500.0	30.0	1.17	54.00	65.0	0.0535	0.0887	0.0928

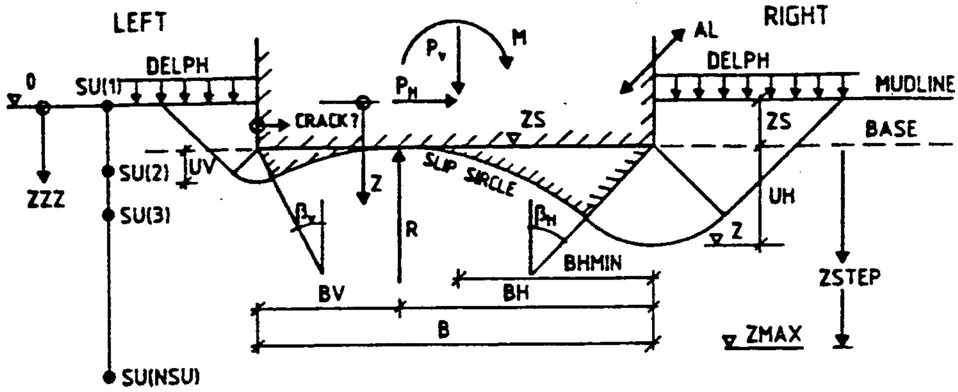
*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

535 0.0887 0.0928

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

STANDARD INPUT DATA FORM FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



PROJECT (1–57)
EXAMPLE 4 – CARL

UNITS USED (1–30)
MN/M

B (1–10)	AL (11–20)	ZSS (21–30)	ZSF (31–40)	ZSSTEP (41–50)	GAMS 1) (51–60)	GAMW 1) (61–70)	PS *) (71–80)
100.0	100.0	30.0	30.0	0.0	0.020	0.010	
100.0	100.0	30.0	30.0	0.0	0.020	0.010	

PVS (1–10)	PVF (11–20)	PVSTEP (21–30)	PH (31–40)	AM (41–50)	DELPH (51–60)	DELPT (61–70)	FAC 2) (71–80)
1500.0	1500.0	0.0	900.0	75000.0	0.005	0.005	1.0
2000.0	2000.0	0.0	750.0	100000.0	0.007	-0.007	1.0

ZMAX (1–10)	ZSTEP (11–20)	REDBS 3) (21–30)	REDSS 4) (31–40)	CRACK 9) (41–50)	PASTYP (51–60)	ZSPEC 1 *) (61–70)	ZSPEC 2 *) (71–80)
70.0	2.0	0.0	0.0	0.0	6.0		
70.0	2.0	0.0	0.0	0.0	6.0		

NSU (1–5)	NPR1 8) (8)	NPR2 8) (11)	NPR3 8) (14)	NPR4 8) (17)	NPR5 8) (20)	BMAX *) (21–30)	DELB *) (31–40)	ALMAX *) (41–50)	DELAL *) (51–60)	RUV 10) (61–65)	RUH 10) (66–70)	BHMIN (71–80)
2	1	1	0	0	0					0.0	0.0	50.0
2	1	1	0	0	0					0.0	0.0	50.0

	ZZZ (1–10)	SU 5) (11–20)	SUD 6) (21–30)	SUP 7) (31–40)	SU2 (41–50)
1	0.0 0.0	0.001 0.150			
2	100.0 100.0	0.225 0.150			
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
   = 1.2, LOAD FACTORS 1.0 AND 1.2
   = -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
   NPR1 = 0 GIVES TABLE OF SF-VALUES
   NPR2 = 0 GIVES ALL K (N) FOR SFMIN
   NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
   NPR4 = 0 OMITTED IN THIS MODE
   NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE = 0. OPEN CRACK.: = 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
*) ZSPEC1 = BH, ZSPEC2 = UH.

*) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.

C A R L

------------------------------------- NGI 1985 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 10 6 1985

PROJECT : EKSEMPEL 4 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M		
WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK
PASSIVE SIDE AREA ASSUMPTION : SMALL AREA

LEFT SIDE ROUGHNESS ASSUMTION : 0.00
RIGHT SIDE ROUGHNESS ASSUMTION : 0.00

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 4 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
32.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
34.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
36.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
38.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
40.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
42.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
44.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
46.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
48.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
50.0	1.77	1.69	1.70	1.75	1.83	1.93	2.05	0.00	0.00	0.00	0.00
52.0	0.00	1.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
54.0	0.00	1.65	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
56.0	0.00	1.65	1.61	1.64	0.00	0.00	0.00	0.00	0.00	0.00	0.00
58.0	0.00	1.66	1.61	1.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00
60.0	0.00	1.67	1.61	1.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00
62.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
64.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
66.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
68.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

MINIMUM SAFETY FACTOR	SF =	1.610
FOUND FOR DEPTH	Z =	58.000
AND DISTANCE TO CIRCLE CENTER	BH =	60.000
AND CIRCLE RADIUS	=	50.286

S(A*SA) =-0.1581E+06
S(B) = 0.9824E+05

PROJECT : EKSEMPEL 4 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

MOMENTS CALCULATED FOR CRITICAL SURFACE
****

MINIMUM SAFETY FACTOR SF = 1.610
FOUND FOR DEPTH Z = 58.000
AND DISTANCE TO CIRCLE CENTER BH = 60.000
AND CIRCLE RADIUS = 50.286

K(N) = A(N) * SA(N)/SF + B(N)

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	MB	0.8700E+05	0.0000E+00	0.000	0.000	0.8700E+05
2	VB	0.7500E+05	0.0000E+00	0.000	0.000	0.7500E+05
3	HB	0.2011E+05	0.0000E+00	0.000	0.000	0.2011E+05
4	PA	0.2811E+05	0.0000E+00	0.035	0.021	0.2811E+05
5	-PP	-0.6108E+05	-0.3629E+06	0.035	0.021	-0.5328E+05
6	T	0.0000E+00	0.0000E+00	0.035	0.021	0.0000E+00
7	HA	0.3936E+05	-0.3132E+06	0.090	0.056	0.5679E+05
8	-HP	-0.1908E+06	-0.7771E+06	0.111	0.069	-0.1371E+06
9	-RP	-0.1934E+05	-0.3907E+06	0.080	0.049	0.0000E+00
10	-RSBS	0.0000E+00	0.0000E+00	0.035	0.021	0.0000E+00
11	-RSSS	0.0000E+00	0.0000E+00	0.062	0.039	0.0000E+00
12	W	0.2163E+05	0.0000E+00	0.000	0.000	0.2163E+05

PROJECT : EKSEMPEL 4 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.00	1500.0	30.0	1.61	58.00	60.0	0.0533	0.0556	0.0573

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

C A R L

---------------------------------- NGI 1985 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 10 6 1985

PROJECT : EKSEMPEL 4 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	750.000
OVERTURNING MOMENT AT MUDLINE	M =	100000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.007
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.007
LOAD FACTORS TO BE USED	LF =	1.0
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK
PASSIVE SIDE AREA ASSUMPTION : SMALL AREA

LEFT SIDE ROUGHNESS ASSUMTION : 0.00
RIGHT SIDE ROUGHNESS ASSUMTION : 0.00

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.150	0.150
100.000	0.150	0.150

PROJECT : EKSEMPEL 4 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 2000.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
32.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
34.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
36.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
38.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
40.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
42.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
44.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
46.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
48.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
50.0	2.00	1.99	2.01	2.07	2.14	2.22	2.31	0.00	0.00	0.00	0.00
52.0	0.00	1.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
54.0	0.00	1.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
56.0	0.00	1.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
58.0	0.00	1.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
60.0	0.00	1.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
62.0	0.00	1.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
64.0	0.00	1.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
66.0	0.00	1.71	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
68.0	0.00	1.70	1.65	1.65	1.68	0.00	0.00	0.00	0.00	0.00	0.00
70.0	0.00	1.70	1.64	1.63	1.66	0.00	0.00	0.00	0.00	0.00	0.00

MINIMUM SAFETY FACTOR SF = 1.633
FOUND FOR DEPTH Z = 70.000
AND DISTANCE TO CIRCLE CENTER BH = 65.000
AND CIRCLE RADIUS = 32.812

S(A*SA) =-0.2641E+06
S(B) = 0.1618E+06

PROJECT : EKSEMPEL 4 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 2000.0 BASE EMB : 30.00

MOMENTS CALCULATED FOR CRITICAL SURFACE
****

MINIMUM SAFETY FACTOR	SF =	1.633
FOUND FOR DEPTH	Z =	70.000
AND DISTANCE TO CIRCLE CENTER	BH =	65.000
AND CIRCLE RADIUS	=	32.812

K(N) = A(N) * SA(N)/SF + B(N)

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	MB	0.1225E+06	0.0000E+00	0.000	0.000	0.1225E+06
2	VB	0.1200E+06	0.0000E+00	0.000	0.000	0.1200E+06
3	HB	0.2461E+05	0.0000E+00	0.000	0.000	0.2461E+05
4	PA	0.2027E+05	0.0000E+00	0.150	0.092	0.2027E+05
5	-PP	-0.6389E+05	-0.2869E+06	0.150	0.092	-0.3753E+05
6	T	0.0000E+00	0.0000E+00	0.150	0.092	0.0000E+00
7	HA	0.1946E+05	-0.2576E+06	0.150	0.092	0.4313E+05
8	-HP	-0.2513E+06	-0.1009E+07	0.150	0.092	-0.1585E+06
9	-RP	-0.1900E+05	-0.2068E+06	0.150	0.092	0.0000E+00
10	-RSBS	0.0000E+00	0.0000E+00	0.150	0.092	0.0000E+00
11	-RSSS	0.0000E+00	0.0000E+00	0.150	0.092	0.0000E+00
12	W	0.2732E+05	0.0000E+00	0.000	0.000	0.2732E+05

PROJECT : EKSEMPEL 4 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

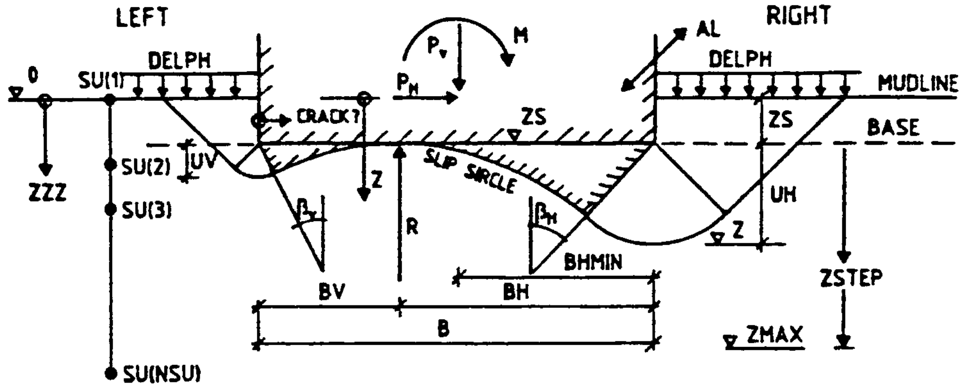
AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.00	2000.0	30.0	1.63	70.00	65.0	0.0919	0.0840	0.0867

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

STANDARD INPUT
DATA FORM
FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



PROJECT (1–57)
EXAMPLE 5 – CARL

UNITS USED (1–30)
MN/M

	1–10	11–20	21–30	31–40	41–50	51–60	61–70	71–80
	B	AL	ZSS	ZSF	ZSSTEP	GAMS 1)	GAMW 1)	PS #)
	100.0	100.0	30.0	30.0	0.0	0.020	0.010	
	PVS	PVF	PVSTEP	PH	AM	DELPH	DELPT	FAC 2)
	1500.0	1500.0	0.0	900.0	75000.0	0.005	-0.005	1.0
	ZMAX	ZSTEP	REDBS 3)	REDSS 4)	CRACK 9)	PASTYP	ZSPEC 1 #)	ZSPEC 2 #)
	70.0	2.0	0.4	0.5	1.0	1.0	75.0	8.0

NSU (1–5)	NPR1 8) (8)	NPR2 8) (11)	NPR3 8) (14)	NPR4 8) (17)	NPR5 8) (20)	BMAX #) (21–30)	DELB #) (31–40)	ALMAX #) (41–50)	DELAL #) (51–60)	RUV 10) (61–65)	RUH 10) (66–70)	BHMIN (71–80)
2	1	1	1	1	0					0.5	0.5	50.0

	ZZZ (1–10)	SU 5) (11–20)	SUD 6) (21–30)	SUP 7) (31–40)	SU2 (41–50)
1	0.0	0.001			//////
2	100.0	0.225			//////
3					//////
4					//////
5					//////
6					//////
7					//////
8					//////
9					//////
10					//////
11					//////
12					//////
13					//////
14					//////
15					//////

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
= 1.2, LOAD FACTORS 1.0 AND 1.2
= -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 = 0 GIVES TABLE OF SF-VALUES
NPR2 = 0 GIVES ALL K (N) FOR SFMIN
NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
NPR4 = 0 OMITTED IN THIS MODE
NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE = 0. OPEN CRACK.:
= 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
#) ZSPEC1 = BH, ZSPEC2 = UH.

#) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
BLANK LINE ENDING ALL DATA SETS.

C A R L

--- NGI 1985 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 10 6 1985

PROJECT : EKSEMPEL 5 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M		
WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.400
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.500

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK
PASSIVE SIDE AREA ASSUMPTION : BIG AREA

LEFT SIDE ROUGHNESS ASSUMTION : 0.50
RIGHT SIDE ROUGHNESS ASSUMTION : 0.50

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 5 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH
****

DEPTH	HOR. DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE										
	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0	90.0	95.0	100.0
32.0	2.45	2.48	2.51	2.54	2.57	2.60	2.62	2.65	2.67	2.69	2.72
34.0	2.15	2.17	2.21	2.25	2.29	2.33	2.37	2.41	2.45	2.49	2.52
36.0	1.96	1.98	2.01	2.05	2.10	2.15	2.19	2.24	2.29	2.33	2.37
38.0	1.84	1.84	1.87	1.91	1.96	2.01	2.06	2.11	2.16	2.21	2.26
40.0	1.76	1.75	1.77	1.81	1.85	1.91	1.96	2.02	2.07	2.13	2.18
42.0	1.72	1.69	1.70	1.73	1.78	1.83	1.88	1.94	2.00	2.06	2.12
44.0	1.70	1.65	1.65	1.68	1.72	1.77	1.83	1.89	1.95	2.01	2.07
46.0	1.69	1.62	1.61	1.64	1.68	1.73	1.79	1.85	1.91	1.97	2.04
48.0	1.70	1.61	1.59	1.61	1.65	1.70	1.76	1.82	1.89	1.95	2.01
50.0	1.72	1.62	1.59	1.60	1.63	1.68	1.74	1.80	1.87	1.94	2.00
52.0	1.76	1.63	1.59	1.59	1.63	1.68	1.73	1.80	1.86	1.93	2.00
54.0	1.80	1.65	1.59	1.59	1.63	1.67	1.73	1.80	1.86	1.93	2.00
56.0	1.85	1.68	1.61	1.60	1.63	1.68	1.74	1.80	1.87	1.94	2.01
58.0	1.91	1.71	1.63	1.62	1.65	1.69	1.75	1.82	1.88	1.96	2.03
60.0	1.98	1.76	1.66	1.64	1.66	1.71	1.77	1.83	1.90	1.98	2.05
62.0	2.05	1.81	1.70	1.67	1.69	1.73	1.79	1.86	1.93	2.00	2.08
64.0	2.14	1.87	1.74	1.70	1.72	1.76	1.82	1.88	1.96	2.03	2.11
66.0	2.23	1.93	1.79	1.74	1.75	1.79	1.85	1.92	1.99	2.07	2.14
68.0	2.33	2.01	1.85	1.79	1.79	1.83	1.89	1.96	2.03	2.11	2.18
70.0	2.44	2.09	1.91	1.84	1.84	1.87	1.93	2.00	2.07	2.15	2.23

```
MINIMUM SAFETY FACTOR              SF =      1.585
FOUND FOR DEPTH                     Z =     50.000
AND DISTANCE TO CIRCLE CENTER      BH =     60.000
AND CIRCLE RADIUS                     =     80.000

S(A*SA)                               =-0.2190E+06
S(B)                                  = 0.1381E+06
```

PROJECT : EKSEMPEL 5 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

MOMENTS CALCULATED FOR CRITICAL SURFACE
****

```
MINIMUM SAFETY FACTOR              SF =      1.585
FOUND FOR DEPTH                     Z =     50.000
AND DISTANCE TO CIRCLE CENTER      BH =     60.000
AND CIRCLE RADIUS                     =     80.000
```

K(N) = A(N) * SA(N)/SF + B(N)

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	MB	0.8700E+05	0.0000E+00	0.000	0.000	0.8700E+05
2	VB	0.7500E+05	0.0000E+00	0.000	0.000	0.7500E+05
3	HB	0.3200E+05	0.0000E+00	0.000	0.000	0.3200E+05
4	PA	0.6831E+05	-0.6466E+06	0.035	0.022	0.8242E+05
5	-PP	-0.9369E+05	-0.6466E+06	0.035	0.022	-0.7957E+05
6	T	-0.2619E+04	-0.1200E+06	0.035	0.022	0.0000E+00
7	HA	0.4049E+05	-0.2798E+06	0.082	0.052	0.5503E+05
8	-HP	-0.1781E+06	-0.6944E+06	0.099	0.062	-0.1348E+06
9	-RP	-0.3486E+05	-0.7086E+06	0.078	0.049	0.0000E+00
10	-RSBS	-0.4982E+04	-0.2282E+06	0.035	0.022	0.0000E+00
11	-RSSS	-0.9654E+04	-0.3235E+06	0.047	0.030	0.0000E+00
12	W	0.2110E+05	0.0000E+00	0.000	0.000	0.2110E+05

SPECIFIED CIRCLE CALCULATION
****

SAFETY FACTOR	SF =	2.01
AT DEPTH	Z =	38.00
HOR. DISTANCE TO CIRCLE CENTER	BH =	75.00
AND CIRCLE RADIUS	R =	347.56

K(N) = A(N) * SA(N)/SF + B(N)

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	MB	0.8700E+05	0.0000E+00	0.000	0.000	0.8700E+05
2	VB	0.1875E+06	0.0000E+00	0.000	0.000	0.1875E+06
3	HB	0.1390E+06	0.0000E+00	0.000	0.000	0.1390E+06
4	PA	0.2830E+06	-0.2565E+07	0.035	0.017	0.3272E+06
5	-PP	-0.3606E+06	-0.2565E+07	0.035	0.017	-0.3164E+06
6	T	-0.2068E+04	-0.1200E+06	0.035	0.017	0.0000E+00
7	HA	0.1625E+05	-0.8483E+05	0.069	0.035	0.1919E+05
8	-HP	-0.2227E+06	-0.8426E+06	0.080	0.040	-0.1893E+06
9	-RP	-0.1245E+06	-0.3435E+07	0.073	0.036	0.0000E+00
10	-RSBS	-0.1488E+05	-0.8635E+06	0.035	0.017	0.0000E+00
11	-RSSS	-0.9775E+04	-0.5592E+06	0.035	0.017	0.0000E+00
12	W	0.2163E+05	0.0000E+00	0.000	0.000	0.2163E+05

PROJECT : EKSEMPEL 5 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

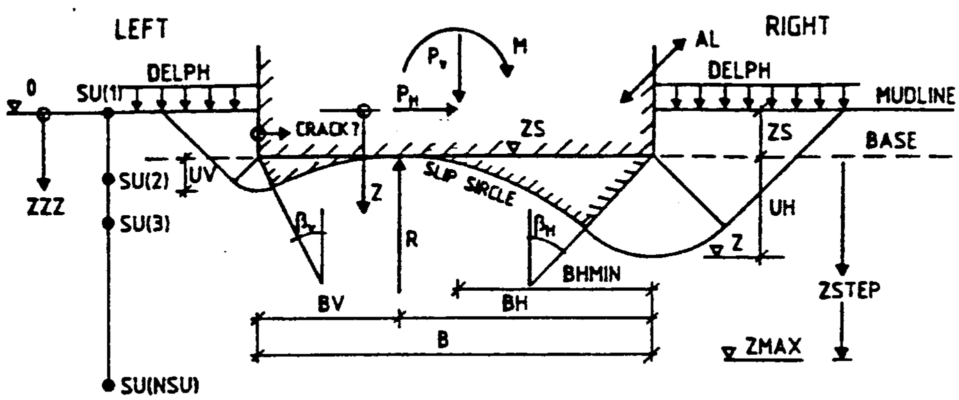
AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.00	1500.0	30.0	1.59	50.00	60.0	0.0379	0.0556	0.0573

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

STANDARD INPUT
DATA FORM
FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



PROJECT (1–57)
EXAMPLE 6 – CARL

UNITS USED (1–30)
MN/M

B (1–10)	AL (11–20)	ZSS (21–30)	ZSF (31–40)	ZSSTEP (41–50)	GAMS 1) (51–60)	GAMW 1) (61–70)	PS *) (71–80)
100.0	100.0	30.0	30.0	0.0	0.020	0.010	

PVS (1–10)	PVF (11–20)	PVSTEP (21–30)	PH (31–40)	AM (41–50)	DELPH (51–60)	DELPT (61–70)	FAC 2) (71–80)
1500.0	1500.0	0.0	400.0	75000.0	0.005	-0.005	1.0

ZMAX (1–10)	ZSTEP (11–20)	REDBS 3) (21–30)	REDSS 4) (31–40)	CRACK 9) (41–50)	PASTYP (51–60)	ZSPEC 1 *) (61–70)	ZSPEC 2 *) (71–80)
70.0	2.0	1.0	1.0	1.0	1.0		

NSU (1–4)	NPR1 8) (5–7)	NPR2 8) (8–10)	NPR3 8) (11–13)	NPR4 8) (14–16)	NPR5 8) (17–20)	BMAX *) (21–30)	DELB *) (31–40)	ALMAX *) (41–50)	DELAL *) (51–60)	RUV 10) (61–65)	RUH 10) (66–70)	BHMIN (71–80)
2	0	1	0	0	0	120.0	10.0	80.0	-10.0	1.0	1.0	50.0

	ZZZ (1–10)	SU 5) (11–20)	SUD 6) (21–30)	SUP 7) (31–40)	SU2 (41–50)
1	0.0	0.001			
2	100.0	0.225			
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
   = 1.2, LOAD FACTORS 1.0 AND 1.2
   = -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
   NPR1 = 0 GIVES TABLE OF SF-VALUES
   NPR2 = 0 GIVES ALL K (N) FOR SFMIN
   NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
   NPR4 = 0 OMITTED IN THIS MODE
   NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE = 0. OPEN CRACK.
   = 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
*) ZSPEC1 = BH, ZSPEC2 = UH.

*) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
BLANK LINE ENDING ALL DATA SETS.

C A R L

------------------------------------ NGI 1985 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 10 6 1985

PROJECT : EKSEMPEL 6 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	1.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK
PASSIVE SIDE AREA ASSUMPTION : BIG AREA

LEFT SIDE ROUGHNESS ASSUMTION : 1.00
RIGHT SIDE ROUGHNESS ASSUMTION : 1.00

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 6 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

MOMENTS CALCULATED FOR CRITICAL SURFACE
****

MINIMUM SAFETY FACTOR SF = 1.851
FOUND FOR DEPTH Z = 50.000
AND DISTANCE TO CIRCLE CENTER BH = 60.000
AND CIRCLE RADIUS = 80.000

K(N) = A(N) * SA(N)/SF + B(N)

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	MB	0.8700E+05	0.0000E+00	0.000	0.000	0.8700E+05
2	VB	0.7500E+05	0.0000E+00	0.000	0.000	0.7500E+05
3	HB	0.3200E+05	0.0000E+00	0.000	0.000	0.3200E+05
4	PA	0.6942E+05	-0.6956E+06	0.035	0.019	0.8242E+05
5	-PP	-0.9258E+05	-0.6956E+06	0.035	0.019	-0.7957E+05
6	T	-0.5607E+04	-0.3000E+06	0.035	0.019	0.0000E+00
7	HA	0.4258E+05	-0.2798E+06	0.082	0.044	0.5503E+05
8	-HP	-0.1719E+06	-0.6944E+06	0.099	0.053	-0.1348E+06
9	-RP	-0.2985E+05	-0.7086E+06	0.078	0.042	0.0000E+00
10	-RSBS	-0.1067E+05	-0.5706E+06	0.035	0.019	0.0000E+00
11	-RSSS	-0.1653E+05	-0.6471E+06	0.047	0.026	0.0000E+00
12	W	0.2110E+05	0.0000E+00	0.000	0.000	0.2110E+05

PROJECT : EKSEMPEL 6 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.00	1500.0	30.0	1.85	50.00	60.0	0.0301	0.0556	0.0573

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI REPORT 52406-44 , DATED 10 06 1985

PROJECT : EKSEMPEL 6 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M		
WIDTH OF FOUNDATION	B =	110.000
LENGTH OF FOUNDATION	L =	90.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	1.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK
PASSIVE SIDE AREA ASSUMPTION : SMALL AREA
ROUGHNESS LEFT SIDE : 1.00
ROUGHNESS RIGHT SIDE : 1.00
DISTANCE TO CIRCLE CENTER : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 6 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

MOMENTS CALCULATED FOR CRITICAL SURFACE
****

MINIMUM SAFETY FACTOR SF = 1.975
FOUND FOR DEPTH Z = 50.000
AND DISTANCE TO CIRCLE CENTER BH = 68.000
AND CIRCLE RADIUS = 105.600

K(N) = A(N) * SA(N)/SF + B(N)

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	MB	0.8700E+05	0.0000E+00	0.000	0.000	0.8700E+05
2	VB	0.9672E+05	0.0000E+00	0.000	0.000	0.9672E+05
3	HB	0.4224E+05	0.0000E+00	0.000	0.000	0.4224E+05
4	PA	0.8118E+05	-0.8037E+06	0.035	0.018	0.9526E+05
5	-PP	-0.1061E+06	-0.8037E+06	0.035	0.018	-0.9201E+05
6	T	-0.5203E+04	-0.2970E+06	0.035	0.018	0.0000E+00
7	HA	0.4317E+05	-0.2642E+06	0.080	0.041	0.5389E+05
8	-HP	-0.1964E+06	-0.7730E+06	0.099	0.050	-0.1578E+06
9	-RP	-0.3775E+05	-0.9541E+06	0.078	0.040	0.0000E+00
10	-RSBS	-0.1395E+05	-0.7964E+06	0.035	0.018	0.0000E+00
11	-RSSS	-0.1906E+05	-0.7909E+06	0.048	0.024	0.0000E+00
12	W	0.2812E+05	0.0000E+00	0.000	0.000	0.2812E+05

PROJECT : EKSEMPEL 6 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.00	1500.0	30.0	1.97	50.00	68.0	0.0280	0.0524	0.0538

*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI REPORT 52406-44 , DATED 10 06 1985

PROJECT : EKSEMPEL 6 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M		
WIDTH OF FOUNDATION	B =	120.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	1.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK
PASSIVE SIDE AREA ASSUMPTION : SMALL AREA
ROUGHNESS LEFT SIDE : 1.00
ROUGHNESS RIGHT SIDE : 1.00
DISTANCE TO CIRCLE CENTER : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 6 ***CARL***
LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

MOMENTS CALCULATED FOR CRITICAL SURFACE
****

MINIMUM SAFETY FACTOR SF = 2.065
FOUND FOR DEPTH Z = 52.000
AND DISTANCE TO CIRCLE CENTER BH = 78.000
AND CIRCLE RADIUS = 127.273

K(N) = A(N) * SA(N)/SF + B(N)

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	MB	0.8700E+05	0.0000E+00	0.000	0.000	0.8700E+05
2	VB	0.1307E+06	0.0000E+00	0.000	0.000	0.1307E+06
3	HB	0.5091E+05	0.0000E+00	0.000	0.000	0.5091E+05
4	PA	0.8633E+05	-0.8482E+06	0.035	0.017	0.1005E+06
5	-PP	-0.1113E+06	-0.8482E+06	0.035	0.017	-0.9713E+05
6	T	-0.4826E+04	-0.2880E+06	0.035	0.017	0.0000E+00
7	HA	0.3862E+05	-0.2264E+06	0.078	0.038	0.4717E+05
8	-HP	-0.2332E+06	-0.8932E+06	0.102	0.049	-0.1892E+06
9	-RP	-0.4307E+05	-0.1126E+07	0.079	0.038	0.0000E+00
10	-RSBS	-0.1723E+05	-0.1028E+07	0.035	0.017	0.0000E+00
11	-RSSS	-0.2386E+05	-0.9872E+06	0.050	0.024	0.0000E+00
12	W	0.3995E+05	0.0000E+00	0.000	0.000	0.3995E+05

PROJECT : EKSEMPEL 6 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE CSS = -S(B)/S(A)
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	DIST. FROM EDGE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M´HOF
LF	PV	ZS	SF	Z	BH	CSS	CB	CM
1.00	1500.0	30.0	2.06	52.00	78.0	0.0272	0.0504	0.0515

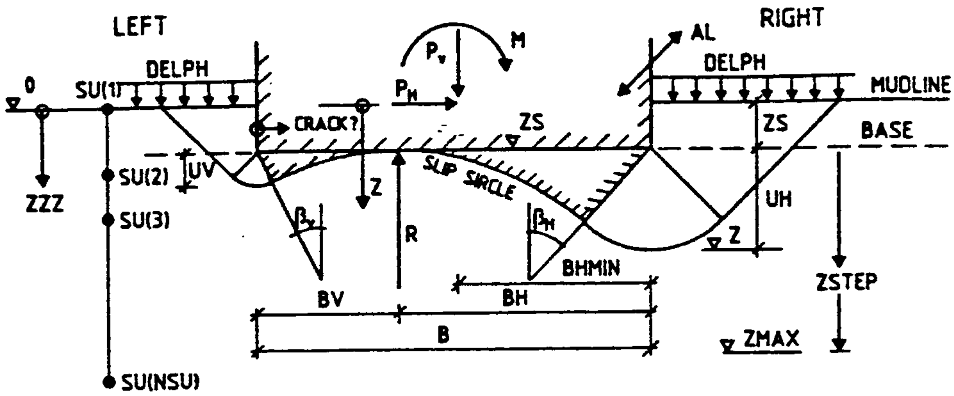
*)
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

NORWEGIAN GEOTECHNICAL INSTITUTE

52406-44-1

STANDARD INPUT
DATA FORM
FOR CARL

BEARING CAPACITY CALCULATED BY SLIP SIRCLES



1–57
PROJECT
EXAMPLE 7 – CARL

1–30
UNITS USED
MN/M

B (1–10)	AL (11–20)	ZSS (21–30)	ZSF (31–40)	ZSSTEP (41–50)	GAMS 1) (51–60)	GAMW 1) (61–70)	PS *) (71–80)
100.0	100.0	30.0	30.0	0.0	0.020	0.010	

PVS (1–10)	PVF (11–20)	PVSTEP (21–30)	PH (31–40)	AM (41–50)	DELPH (51–60)	DELPT (61–70)	FAC 2) (71–80)
2000.0	2000.0	0.0	750.0	100000.0	0.070	0.070	1.3

ZMAX (1–10)	ZSTEP (11–20)	REDBS 3) (21–30)	REDSS 4) (31–40)	CRACK 9) (41–50)	PASTYP (51–60)	ZSPEC 1 *) (61–70)	ZSPEC 2 *) (71–80)
70.0	2.0	0.5	0.4	1.0	1.0		

NSU (1–5)	NPR1 8) (6–8)	NPR2 8) (9–11)	NPR3 8) (12–14)	NPR4 8) (15–17)	NPR5 8) (18–20)	BMAX *) (21–30)	DELB *) (31–40)	ALMAX *) (41–50)	DELAL *) (51–60)	RUV 10)	RUH 10)	BHMIN (71–80)
2	1	1	1	1	1					0.5	0.5	50.0

	ZZZ (1–10)	SU 5) (11–20)	SUD 6) (21–30)	SUP 7) (31–40)	SU2 (41–50)
1	0.0	0.001	0.001	0.001	0.001
2	100.0	0.225	0.200	0.180	0.050
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT
   = 1.2, LOAD FACTORS 1.0 AND 1.2
   = -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
   NPR1 = 0 GIVES TABLE OF SF-VALUES
   NPR2 = 0 GIVES ALL K (N) FOR SFMIN
   NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
   NPR4 = 0 OMITTED IN THIS MODE
   NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON LEFT SIDE. = 0. OPEN CRACK.: = 1.0 CLOSED CRACK.
10) EARTH PRESSURE COEFFICIENTS ON LEFT AND RIGHT SIDE.
*) ZSPEC1 = BH, ZSPEC2 = UH.

*) MAY BE LEFT BLANK.
//// INPUT DATA OMITTED IN THIS FAILURE MODE.
LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
BLANK LINE ENDING ALL DATA SETS.

C A R L

------------------------------------ NGI 1985 ----

BEARING CAPACITY CALCULATED BY SLIP CIRCLES PAGE 1

METHOD DESCRIBED IN NGI REPORT 52406-44 DATED 10 6 1985

PROJECT : EKSEMPEL 7 ***CARL***

INPUT DATA REPRINT
****

UNITS USED : SI-ENHETER DVS MN & M		
WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010
SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	750.000
OVERTURNING MOMENT AT MUDLINE	M =	100000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.070
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.070
LOAD FACTORS TO BE USED	LF =	1.3
MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.500
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK
PASSIVE SIDE AREA ASSUMPTION : BIG AREA

LEFT SIDE ROUGHNESS ASSUMTION : 0.50
RIGHT SIDE ROUGHNESS ASSUMTION : 0.50

MINIMUM DISTANCE FROM CIRCLE CENTRE TO RIGHT EDGE OF BASE : 50.0

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EKSEMPEL 7 ***CARL***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS
****

MINIMUM HORIZONTAL FORCE WHEN THE
SAFETY FACTOR IS SET TO 1.0

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	RADIUS OF CIRCLE	-DEPTH SLIP SURFA	DIST. FROM EDGE	MIN HOR FORCE
LF	PV	ZS	R	Z	BH	PH
1.00	2000.0	30.0	70.8	52.0	60.0	155.9